

Introduction to computing

C/C++ language :-

↳ they are medium level language.

polyglot programmer



know almost

5-6 languages etc.

C/C++ → compile language → source code → machine code

genu C compiler / genu C++ compiler

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    cout << "Hello world" << endl;
```

```
    return 0;
```

```
}
```

```
void if_test() {
```

```
    int x = 29;
```

```
    if (x < 25) {
```

```
        cout << "x is less than 25" << endl;
```

```
    } else {
```

```
        cout << "x is not less than 25" << endl;
```

```
        return;
```

```
    }
```


cout << <sup>thing that run
on console</sup>

return <sub>value that
return on call</sub>

#booleans

```
int v=0;
```

```
if (v)
```

```
    cout << "v is true! << endl;
```

```
else
```

```
    cout << "v is false" << endl;
```

(0 is false all other are true)

// boolean operator

and true && true = 1

or true || false = 1

not ! true = 0

// ternary operator

for true
for false

syntax

$A ? B$

int t = (3 > 2) ? 7 : 3;

A

B

C

$A ? C : B$;

must be
conditional

cout << "t = " << t << endl;

// iteration

void while test() {

int i = 0;

while (i < 5) {

cout << i << endl;

i++; > it is not in python

}

it is

this is not recommended

bad
practice

use $i = i + 1$, $i += 1$;

// array : ^{they python}
list and C/C++ array are differ

```
void array_test() {
```

```
// collection of element of same type with fixed  
size not have append operationin it (you not add  
more values)
```

```
// list is expandable but array is not but they  
have same shape.
```

```
// list are very useful in industry  
int a[5] = {};
```

```
for (int i = 0; i < 5; i++) {  
    cout << a[i] << " "  
}
```


int a[2][3]]-> 2-D array.
 ↓ ↓
 row cols

0 0 0

0 0 0

```
for (int i = 0; i < 2; i++) {  
    for (int j = 0; j < 3; j++) {
```

```
        cout << a[i][j] << " ";
```

```
    }
```

```
    cout << endl;
```

```
}
```

2-D array

// structure it is a kind of class, a template

```
struct student {  
    int rollno;  
    float marks;  
};
```

data type variable

student s1;

student s2;

s1.rollno = 19

s1.marks = 84.4

s2.rollno = 20

s2.marks = 90

How to code :-

```
#include <iostream>
using namespace std;
```

} - 1-step

```
void array_test {
```

```
    cout << "testing... " << endl;
```

```
}
```

} -> 3-step

```
int main() {
```

```
    array_test();
```

```
    return 0;
```

```
}
```

} -> 2-step